

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF VERIFICATION****Note:** This table completed by ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Ducted Cooling System Information

01	Space Conditioning System Identification or Name	
02	Space Conditioning System Description of Area Served	
03	Indoor Unit Name	
04	System Installation Type	
05	Nominal Cooling Capacity (tons)	
06	Condenser Speed Type	
07	Cooling System Zonal Control Type	
08	Central Fan Integrated (CFI) Ventilation System Status	
09	System Bypass Duct Status	
10	Date of System Airflow Rate Measurement	
11	Airflow Rate Protocol Utilized	
12	Central Fan Ventilation Cooling System Status	

B. Hole for the Placement of a Static Pressure Probe (HSPP), and Permanently Installed Static Pressure Probe (PSPP) in the Supply Plenum

Procedures for installing HSPP or PSPP are specified in RA3.3.1.1.

01	Method Used to Demonstrate Compliance with the HSPP/PSPP Requirement	
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C. Airflow Rate Measurement Apparatus and Procedure Information

Instrument Specifications are given in RA3.3.1.1, and system airflow rate measurement apparatus information is given in RA3.3.2.

01	Airflow Rate Measurement Type Used for this Airflow Rate Verification	
02	Manufacturer of Airflow Measurement Apparatus	
03	Model number of Airflow Measurement Apparatus	
04	Certification Status of the Airflow Measurement Apparatus Accuracy	

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****D. Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

The installer shall attempt to correct non-compliant system airflow rates by performing the following remedial actions as specified in RA3.3.3.1.5

01	Determine that the air filter media is clean. If the air filter media is dirty, then replace it with clean filter media.	
02	Open all registers and dampers and remove any obstructions.	
03	Replace/Repair all accessible crushed, blocked, restricted, remove excess length, and sharp bends in ducts. Supported every 4 ft max. with a max. 2 in sag.	
08	Clean the evaporator coil according to the manufacturer and ensure the coil is not obstructed.	
05	Air handler fan speed set to high and blower wheel and motor are operating properly.	
06	If determined to be too small, replace the return duct with a larger one and/or add a second return duct.	
07	If determined to be too small, replace the return grille with a larger area grille.	
08	If any of the above were not completed list the Action Required and a description of why the action could not be completed:	
09	Verification Status:	
10	Correction Notes:	
11	Optional Notes:	

E1. Forced Air System Airflow Rate Measurement

Complete this section with Airflow Rate Measurement Data for Newly Installed Non-Zoned Systems or Zoned Multi-Speed Compressor

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.		
01	Actual System Airflow Rate Measurement (cfm)	
02	Required Minimum System Airflow Rate (cfm/ton)	
03	Required Minimum System Airflow Target (cfm)	
04	Compliance Statement:	

E2. Forced Air System Airflow Rate Measurement - All Zones Calling

Complete this section with Airflow Rate Measurement Data for Newly Installed Zoned Single-Speed Compressor Systems

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.		
01	Actual System Airflow Rate Measurement (cfm)	
02	Required All Zones Calling Minimum System Airflow Rate (cfm/ton)	
03	Required All Zones Calling Minimum System Airflow Target (cfm)	
04	Compliance Statement:	

E3. Forced Air System Airflow Rate Measurement - Best Airflow Rate Attainable

Complete this section with Airflow Rate Measurement Data for Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.		
01	Actual System Airflow Rate Measurement (cfm)	
02	Required Minimum System Airflow Rate (cfm/ton)	
03	Required Minimum System Airflow Target (cfm)	
04	Compliance Statement:	
05	ECC Sample Group Eligibility	

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****E4. Forced Air System Airflow Rate Measurement – Heating Only**

Complete this section with Airflow Rate Measurement Data for Newly Installed Heating Only Non-Zoned Systems or Zoned Multi-Speed Compressor Measurement Only – No Minimum Target Requirement

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.		
01	Actual System Airflow Rate Measurement (cfm)	

F. Forced Air System Airflow Rate Measurement – All Other Zonal Control Modes

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3. For compliance with verification in all zonal control modes, it is sufficient to verify airflow rate for operation of each individual zone when the individual zone is the sole zone calling for conditioning. It is not necessary to verify airflow rate for combinations of 2 or more zones that are less than all zones calling (e.g., 2 out of three zones calling).

01	Number of Independently Controlled Zones (i.e., number of thermostats or temperature sensors that independently control one or more dampers.)		
02	Required Minimum Cooling System Airflow Rate (cfm/ton)		
03	Required Minimum Airflow in all Zonal Control Modes (cfm)		
04		05	06
Zone Name		Zone Description	Measured Airflow with All Other Zones Off (CFM)
			Zone Compliance Status
08	Compliance Statement:		

G. Central Fan Ventilation Cooling System Airflow Rate Measurement

The procedures for central fan ventilation cooling system airflow rate verification are specified in Reference Residential Appendix RA3.3.4

01	Required Ventilation System Airflow Rate (cfm)	
02	Actual System Ventilation Airflow Rate Measurement (cfm)	
03	Compliance Statement:	

H1. Additional Requirements for All Systems

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Air filters that meet the applicable requirements of Standards Section 150.0(m)12 or 150.0(m)13 were properly installed in the system during system airflow rate measurement identified on this Certificate of Installation.	
02	The airflow rate measurement apparatus used to perform the airflow rate measurement identified on this Certificate of Installation was calibrated in accordance with the apparatus manufacturer's specifications and conforms to the instrumentation specifications given in RA3.3.1.	
03	All registers were fully open during the diagnostic test.	
04	System fan was set at maximum speed during the diagnostic test.	
05	If fresh air duct is part of the HVAC system it was not closed during the diagnostic test.	
06	Airflow rate and fan watt draw shall be simultaneous measurements when used to calculate the Fan Efficacy tested value.	
07	Verification Status:	
08	Correction Notes:	

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****H2. Additional Requirements for Systems with Heating And Cooling**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	A visual inspection shall confirm that bypass ducts that deliver conditioned supply air directly to the space conditioning system return duct airflow are not used on newly constructed zonally controlled systems unless the Performance Certificate of Compliance indicates an allowance for use of a bypass duct. When a bypass duct is accounted for on the Performance Certificate of Compliance, the airflow rate shall conform to the specifications listed on the Certificate of Compliance.	
02	Multi-speed compressor space cooling systems or variable speed compressor systems with controls that vary fan speed subject to the number of zones, as certified by the installer may verify airflow (cfm/ton) and fan efficacy (Watt/cfm) with system operating at maximum compressor capacity and system fan speed with all zones calling for conditioning,	
03	Verification Status:	
04	Correction Notes:	

H3. Additional Requirements for Altered Systems Using Alternative Compliance

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	For altered systems that do not comply with the minimum 300 cfm per ton airflow rate requirement but opt to comply using the remedial actions on this MCH-23 compliance document according to Section RA3.3.3.1.5, the system's thermostat shall conform to the specifications in Reference Joint Appendix JA5 and shall be capable of receiving and responding to Demand Response Signals prior to final approval of the building permit by the enforcing agency (Section 150.2(b)1Fiia).	
02	Verification Status:	
03	Correction Notes:	

I. Determination of ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****DOCUMENTATION AUTHOR'S DECLARATION STATEMENT**

1. I certify that this Certificate of Verification documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Date Signed:
Address:	CEA/ECC Certification Information (if applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified ECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require Field verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (CF2R) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (CF1R) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION

Company Name (Installing Subcontractor, General Contractor, or Builder/Owner):	
Responsible Builder or Installer Name:	CSLB License:

ECC PROVIDER DATA REGISTRY INFORMATION

Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable):
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ECC RATER INFORMATION

ECC Rater Company Name:	
Responsible Rater Name:	Responsible Rater Signature:
Responsible Rater Certification Number w/ this ECC Provider:	Date Signed:

CF2R-MCH-23-H User Instructions

Section A. Ducted Cooling System Information

1. System Identification or Name: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.
2. System Location or Area Served: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.
3. Indoor Unit Name: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.
4. System Installation Type: Select the appropriate System Installation Type from the following choices:
 - a. New: Use this choice for newly constructed buildings, additions with all-new systems dedicated to the addition, or new systems installed in existing homes where the equipment and ducts are all newly installed (aka, "Cut-in").
 - b. Replacement: Use this choice if the system is a complete replacement space-conditioning system installed as part of an alteration, and includes all the system heating or cooling equipment plus a replacement duct system (150.2(b)1Diia) where the ducts are at least 75% or more newly installed duct material (up to 25% of the finished system may consist of reused parts from the dwelling unit's previously existing duct system, such as registers, grilles, boots, air handler, coil, plenums, duct material); plus a replacement air handler.
 - c. Alteration: Use this choice for existing buildings where any of the following are newly installed or replaced as part of the project and the system does not meet one of the other compliance categories above.:
 - i. 40 feet or more of space-conditioning system ducts are installed in unconditioned space or indirectly conditioned space.
 - ii. Air conditioning or heat pump condenser
 - iii. Heating or cooling coil
 - iv. Air handler (e.g., furnace, fan coil, package unit)
5. Nominal Cooling Capacity (tons): This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document. If the number of indoor units connected to the outdoor unit is equal to one or the system is a packaged system then this field is equal to the nominal cooling capacity of the condenser. If the number of indoor units connected to the outdoor unit is greater than one this field is equal to the indoor unit nominal cooling capacity.
6. Condenser Speed Type: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.
7. Cooling System Zonal Control Type: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.
8. Central Fan Integrated (CFI) Ventilation System Status: If the system has Central Fan Integrated System, then select "CFI System", otherwise select "Not a CFI system".
9. System Bypass Duct Status: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.
10. Date of System Airflow Rate Measurement: Enter the date that the airflow test was performed.
11. Airflow Rate Protocol Utilized: If the system installation type is "New" or "Replacement" then only the RA3.3 airflow methods may be used. If the system installation type is "Alteration", the RA3.3 airflow methods may be used, but the Alternative to Compliance with Minimum System Airflow Requirements ("Best I Can Do" Airflow) is an option for existing systems that may require substantial modification to improve the airflow.

12. Central Fan Ventilation Cooling System (CFVCS) Status: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document.

Section B. Hole for the Placement of a Static Pressure Probe (HSPP), and Permanently Installed Static Pressure Probe (PSPP) in the Supply Plenum

1. A hole for a static pressure probe (HSPP) or a permanent static pressure probe (PSPP) is required when system airflow verification is required, whether the airflow test method used requires one or not. Select the appropriate choice from the following options using a dropdown box, the Static Pressure Measurement Method:
 - If an Hole Static Pressure Probe is installed then select “HSPP Installed”
 - If a Permanent Static Pressure Probe is installed then select “PSPP Installed”
 - If the system is configured such that an HSPP nor PSPP can be installed, an alternate location that provides access for making supply plenum pressure measurement may be used. Select “An alternative location has been provided and clearly labeled.”
 - If the system is such that an HSPP or PSPP is not applicable, select “HSPP/PSPP are not applicable to this system”.

Section C. Airflow Rate Measurement Apparatus and Procedure Information

1. Airflow Rate Measurement Type Used for this Airflow Rate Verification: Select the appropriate airflow test procedure from the following options for the method used to determine actual fan airflow:
 - Diagnostic Fan Flow Using Fan Flow Meter (aka Plenum Pressure Matching) according to the procedures in RA3.3.3.1.1
 - Diagnostic Fan Flow Using Flow Grid Measurement according to the procedures in RA3.3.3.1.2
 - Diagnostic Fan Flow Using Powered Flow Capture Hood according to the procedures in RA3.3.3.1.3
 - Diagnostic Fan Flow Using Traditional Flow Capture Hood according to the procedures in RA3.3.3.1.4
2. Manufacturer of Airflow Measurement Apparatus: Enter the name of the manufacturer of the airflow measurement tool used to measure the airflow for this test.
3. Model number of Airflow Measurement Apparatus: Enter the model number of the airflow measurement tool used to measure the airflow for this test.
4. Certification Status of the Airflow Measurement Apparatus Accuracy: The measurement apparatus used to perform an airflow verification measurements must appear on the CEC list of approved devices found at http://www.energy.ca.gov/title24/equipment_cert/ama_fas/index.html, if this is true, select “Certified”, otherwise select “Not Certified”. The latter choice will not allow the system to pass until a certified device is used.

D. Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems

(This section is required for altered systems using alternative compliance)

These fields are required for alteration project compliance

1. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.
2. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.
3. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.
4. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.
5. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.
6. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.

7. Refer to section RA3.3.3. for details on this item. Indicate whether completed or not.
8. User Input
9. *Verification Status*: If this Section does not apply, then select “All N/A”. If the system meets the airflow criteria, then select “Pass”, otherwise select “Fail”. The latter selection means that the system does not meet the requirements and the CF1R will have to be revised, or the system will need to be modified to meet the requirements.
10. *Correction Notes*: If one or more applicable requirements are not met “Fail” will appear in the row above. When this occurs, the rater is required to enter detailed notes here that describe what failed and why.
11. If any of the above items could not be completed due to inaccessibility or significant cost, provide additional explanation here.

Section E1. Forced Air System Airflow Rate Measurement

(This section is required for non-zoned systems or zoned systems with multi-speed systems)

1. Required Minimum System Airflow Rate (cfm/ton): This field is filled automatically. The target is based on whether the system is new or altered and whether a value was specified on the CF2R-MCH-01.
2. Required Minimum System Airflow Target (cfm): This field is calculated automatically. It is the product of the minimum airflow rate per ton and the tonnage of the system condenser.
3. Actual System Airflow Rate Measurement (cfm): Enter the actual tested value of the airflow measured using the apparatus specified above.
4. Compliance Statement: This field is filled automatically. Compliance requires that the measured airflow meets the minimum airflow target.

Section E2. Forced Air System Airflow Rate Measurement – All Zones Calling (This section is required if system is zonally controlled)

1. Required All Zones Calling Minimum System Airflow Rate (cfm/ton): This field is filled automatically. The target is based on whether the system is new or altered and whether a value was specified on the CF2R-MCH-01.
2. Required All Zones Calling Minimum System Airflow target (cfm): This field is calculated automatically. It is the product of the minimum airflow rate per ton and the tonnage of the system condenser.
3. Actual System Airflow Rate Measurement (cfm): Enter the actual tested value of the airflow measured using the apparatus specified above.
4. Compliance Statement: This field is filled automatically. Compliance requires that the measured airflow meets the minimum airflow target.

Section E3. Forced Air System Airflow Rate Measurement - Best Airflow Rate Attainable (This section is required for altered systems using alternative compliance)

1. Required Minimum System Airflow Rate (cfm/ton): This field is filled automatically. The target is always 300 cfm/ton for this option.
2. Required Minimum System Airflow Target (cfm): This field is calculated automatically. It is the product of the minimum airflow rate per ton and the tonnage of the system condenser.
3. Actual System Airflow Rate Measurement (cfm): Enter the actual tested value of the airflow measured using the apparatus specified above.
4. Compliance Statement: This field is filled automatically. Compliance requires that the measured airflow meets the minimum airflow target, however if the criteria of RA3.3.3 is met the best attainable airflow rate will suffice.

5. ECC Sample Group Eligibility: This field is filled out automatically. If the minimum airflow rate cannot be met and the criteria of RA3.3.3 is used, the system cannot be included in a ECC sample group.

Section E4. Forced Air System Airflow Rate Measurement

(This section is required if system is heating only)

- Actual System Airflow Rate Measurement (cfm): Enter the actual tested value of the airflow measured using the apparatus specified above.

Section F. Forced Air System Airflow Rate Measurement – All Other Zonal Control Modes

(This section is required for systems with single speed compressor and zonally controlled).

1. Number of Independently Controlled Zones: Enter the number of zones in this system that are independently controlled, i.e., that can call for cooling while other zones can be fully or mostly shut off from system airflow. This usually corresponds to the number of thermostats or zone sensors.
2. Required Minimum Cooling System Airflow Rate (cfm/ton): This field is filled automatically. The target is based on whether the system is new or altered and whether a value was specified on the CF2R-MCH-01.
3. Required Minimum Airflow in all Zonal Control Modes (cfm): This field is filled out automatically. If a value other than 350 cfm was claimed in the performance calculations, it will be referenced from the CF1R, otherwise the target is 350 cfm.
4. Zone Name: Enter a unique name for each zone on this system. Examples: Zone 1, Z1, Zone A, etc.
5. Zone Description: Enter a brief description of each zone that is detailed enough allow someone to distinguish it from the others in the field. Examples: upstairs, first floor, east wing, bedrooms only, (list rooms served), etc.
6. Measured Airflow with All Other Zones Off: This test must be performed with only one independently controlled zone calling for cooling (Note: if fan watt verification is required, it must be performed simultaneously to the corresponding airflow from this test). All other zones must not be calling during this test. The zone dampers for the other zones must be in their normal closed position. Enter the airflow value measured for the zone that is calling. This test must be performed for each and every independently controlled zone.
7. Zone Compliance Status: This field is filled out automatically. The result is based on whether or not the actual airflow meets the required airflow for this zone.
8. Compliance Statement: This field is filled out automatically. The result is based on whether or not the actual airflow meets the required airflow for all zones

Section G. Central Fan Ventilation Cooling System Airflow Rate Measurement

(This section is required if project has a fixed or variable CFVCS.)

1. Required Ventilation System Airflow Rate (cfm): This field is filled automatically. The target is based on the airflow rate specified on the CF2R-MCH-01.
2. Actual System Ventilation Airflow Rate Measurement (cfm): Enter the actual tested value of the airflow measured using the apparatus specified above.
3. Compliance Statement: This field is filled automatically. Compliance requires that the measured airflow meets the airflow target.

Section H1. Additional Requirements for All Systems

1. This field must be a true statement (or not applicable) for the system to comply.
2. This field must be a true statement (or not applicable) for the system to comply.
3. This field must be a true statement (or not applicable) for the system to comply.
4. This field must be a true statement (or not applicable) for the system to comply.
5. This field must be a true statement (or not applicable) for the system to comply.
6. This field must be a true statement (or not applicable) for the system to comply.
7. *Verification Status:* If this Section does not apply, then select “All N/A”. If the system meets the airflow criteria then select “Pass”, otherwise select “Fail”. The latter selection means that the system does not meet the requirements and the CF1R will have to be revised, or the system will need to be modified to meet the requirements.
8. *Correction Notes:* If one or more applicable requirements are not met “Fail” will appear in the row above. When this occurs the rater is required to enter detailed notes here that describe what failed and why.

Section H2. Additional Requirements for Systems with Heating And Cooling

(If project is not heating only, this section is required)

1. This field must be a true statement (or not applicable) for the system to comply.
2. This field must be a true statement (or not applicable) for the system to comply.
3. *Verification Status:* If this Section does not apply, then select “All N/A”. If the system meets the airflow criteria then select “Pass”, otherwise select “Fail”. The latter selection means that the system does not meet the requirements and the CF1R will have to be revised, or the system will need to be modified to meet the requirements.
4. *Correction Notes:* If one or more applicable requirements are not met “Fail” will appear in the row above. When this occurs the rater is required to enter detailed notes here that describe what failed and why.

Section H3. Additional Requirements for Altered Systems Using Alternative Compliance

(If project is an alteration complying with RA3.3.3.1.5, this section is required)

1. This field must be a true statement (or not applicable) for the altered system using alternative compliance to comply.
3. *Verification Status:* If this Section does not apply, then select “All N/A”. If the system meets the airflow criteria then select “Pass”, otherwise select “Fail”. The latter selection means that the system does not meet the requirements and the CF1R will have to be revised, or the system will need to be modified to meet the requirements.
4. *Correction Notes:* If one or more applicable requirements are not met “Fail” will appear in the row above. When this occurs the rater is required to enter detailed notes here that describe what failed and why.

A. Ducted Cooling System Information

01	Space Conditioning System Identification or Name	<<auto filled text: referenced from MCH01>>
02	Space Conditioning System Description of Area Served	<<auto filled text: referenced from MCH01>>
03	Indoor Unit Name	<<auto filled text: referenced from MCH01>>
04	System Installation Type	<<if the parent is a MCH-01b and a “yes” answer was given in B08 or B09, then User pick from list: *New; or *Replacement; else User pick from complete list: *New; or *Replacement; or *Alteration)>>
05	Nominal Cooling Capacity (tons)	<<calculated field: The assigned result value cannot be overwritten by the user except for MCH23a, MCH23b, MCH23c, MCH23e, and MCH23f if parent is MCH01d and specific condition allows it. If Cooling System Type on MCH-01 = ‘NoCooling’, then, result in this field = ‘N/A - Heating-only system’; Elseif, parent is MCH-01a evaluate these fields from the Mch01a: if the duct system in M03 has a value in field D06 > 1, and the system type in fields D04 or D05 are one of the following three types: 1:[VCHP-Ducted], 2:[VCHP-Ductless], 3:[VCHP-Ducted+Ductless], then value in this field = Indoor Unit Nominal Cooling Capacity value referenced field G10, else value in this field = Condenser Nominal Cooling Capacity value referenced from one of the following two locations: 1: field F09 (if SC system is not a HP)] 2: field I10 (if SC system is a HP)] Elseif parent is MCH-01b evaluate these fields from the Mch01b: if the duct system in field J03 has a value in field C12 > 1, and the system type in C03 or C07 are one of the following three types: 1:[VCHP-Ducted], 2:[VCHP-Ductless], 3:[VCHP-Ducted+Ductless], then value in this field = Indoor Unit Nominal Cooling Capacity value referenced from one of the following two locations: 1: field F12 (if < 75% of duct system altered)] 2: field G15 (if new or completely replaced duct system)] else value in this field = Condenser Nominal Cooling Capacity value referenced from field E09 Elseif parent is MCH-01c evaluate these fields from the Mch01c: if the duct system in field L03 has a value in field C11 > 1, and the system type in fields C04 or C05 are one of the following three types; 1:[VCHP-Ducted], 2:[VCHP-Ductless], 3:[VCHP-Ducted+Ductless], then value in this field = Indoor Unit Nominal Cooling Capacity value reference from field F10 else value in this field = Condenser Nominal Cooling Capacity value referenced from one of the following two locations: 1: field E09 (if SC system is not a HP)] 2: field H10 (if SC system is a HP)] Elseif parent is MCH-01d evaluate these fields from the Mch01d: if the duct system in field O03 has a value in field D06 > 1, and the system type in fields D04 or D05 are one of the following three types: 1:[VCHP-Ducted], 2:[VCHP-Ductless], 3:[VCHP-Ducted+Ductless], then value in this field = Indoor Unit Nominal Cooling Capacity value referenced from one of the following two locations: 1: field K12 (if < 75% of duct system altered)] 2: field L15 (if new or completely replaced duct system)] else value in this field = Condenser Nominal Cooling Capacity value referenced from one of the following two locations: 1: field G09 (if SC system is not a HP)] 2: field J10 (if SC system is a HP)]; Else If D05_ResidentialCoolingSystemType = NoCooling And value could not be found in parent MCH01d, allow user input End If >>

06	Condenser Speed Type	<<calculated field: If cooling system type=NoCooling then display text result= n/a - Heating-only system; elseif the condenser speed type is not given on the MCH-01, then display text n/a –Condenser Speed requirements do not apply; elseif the condenser speed type is given on the MCH-01, then display value for Condenser Speed Type (note on applicable MCH-01 docs the options are: MultiSpeed; or SingleSpeed)>>
07	Cooling System Zonal Control Type	<<calculated field: If cooling system type=NoCooling then display text result= n/a - Heating-only system; elseif the Cooling System Zonal Control Type is not given on the MCH-01, then display text: n/a –Cooling System Zonal Control requirements do not apply; elseif the Cooling System Zonal Control Type is given on the MCH-01, then display value for Cooling System Zonal Control Type (note on applicable MCH-01 docs the options are: ZonallyControlled and NotZonal)>>
08	Central Fan Integrated (CFI) Ventilation System Status	<<calculated field: reference value from MCH-01; note: allowable values = *CFI System *Not CFI>>
09	System Bypass Duct Status	<<calculated field: If parent is MCH-01b, MCH-01c, or MCH-01d then value=N/A; elseif parent is MCH-01a, reference value from MCH-01a J10; (note: on the MCH01a the options are: Has Bypass Duct; or No bypass duct)>>
10	Date of System Airflow Rate Measurement	<<user input: date: use validated date format>>
11	Airflow Rate Protocol Utilized	<<Calculated field: If A04=(New or Replacement), then display result = RA3.3 procedures for airflow rate measurement; elseif A04=Alteration then allow user to pick from list: **RA3.3.3.1.5 Alternative to Compliance with Minimum System Airflow Requirements, or ** RA3.3 procedures for airflow rate measurement>>
12	Central Fan Ventilation Cooling System Status	<<Calculated Field: Referenced from MCH-01, if MCH-01 variant is b or c, then display 'Not a CFVCS', Else If MCH-01 variant is a or d If Mch01 B05 Central Fan Ventilation Cooling System Type = Variable, then display 'Variable CFVCS', Or Fixed, then display 'Fixed CFVCS', Else display 'Not a CFVCS'.>>

B. Hole for the Placement of a Static Pressure Probe (HSPP), and Permanently Installed Static Pressure Probe (PSPP) in the Supply Plenum
Procedures for installing HSPP or PSPP are specified in RA3.3.1.1.

01	Method Used to Demonstrate Compliance with the HSPP/PSPP Requirement	<<user select one of the options from list: **HSPP installed and labeled consistent with Figure RA3.3-1; or **PSPP installed and labeled consistent with Figure RA3.3-1, or **HSPP/PSPP cannot be installed consistent with Figure RA3.3-1. An alternative location has been provided and clearly labeled, or ** HSPP/PSPP are not applicable to this system >>
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C. Airflow Rate Measurement Apparatus and Procedure Information

Instrument Specifications are given in RA3.3.1.1, and system airflow rate measurement apparatus information is given in RA3.3.2.

01	Airflow Rate Measurement Type Used for this Airflow Rate Verification	<< user select one from list: Fan Flowmeter according to procedure in RA3.3.3.1.1 Flow Grid according to procedure in RA3.3.3.1.2 Powered Flow Capture Hood according to procedure in RA3.3.3.1.3 Traditional Flow Capture Hood according to procedure in RA3.3.3.1.4>>
02	Manufacturer of Airflow Measurement Apparatus	<<user entry : text (or consider referencing from a data registry user profile)>>
03	Model number of Airflow Measurement Apparatus	<<user entry: text (or consider referencing from a data registry user profile)>>
04	Certification Status of the Airflow Measurement Apparatus Accuracy	<<user select from list: 2. Certified by Manufacturer and listed on CEC Website at http://www.energy.ca.gov/title24/equipment_cert/ama_fas/index.html , or Not Certified (do not continue)>>

D. Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems

<< If A04 = Altered And A11 = RA3.3.3.1.5 this section is required Else only display Section title and this note:

This section does not apply to this project.>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

The installer shall attempt to correct non-compliant system airflow rates by performing the following remedial actions as specified in RA3.3.3.1.5

01	Determine that the air filter media is clean. If the air filter media is dirty, then replace it with clean filter media.	<<user input, pull down list: Completed; Not Completed>>
02	Open all registers and dampers and remove any obstructions.	<<user input, pull down list: Completed; Not Completed>>
03	Replace/Repair all accessible crushed, blocked, restricted, remove excess length, and sharp bends in ducts. Supported every 4 ft max. with a max. 2 in sag.	<<user input, pull down list: Completed; Not Completed>>
08	Clean the evaporator coil according to the manufacturer and ensure the coil is not obstructed.	<<user input, pull down list: Completed; Not Completed>>
05	Air handler fan speed set to high and blower wheel and motor are operating properly.	<<user input, pull down list: Completed; Not Completed>>
06	If determined to be too small, replace the return duct with a larger one and/or add a second return duct.	<<user input, pull down list: Completed; Not Completed>>
07	If determined to be too small, replace the return grille with a larger area grille.	<<user input, pull down list: Completed; Not Completed>>
08	If any of the above were not completed list the Action Required and a description of why the action could not be completed:	<<user input, text, maximum 500 characters>>
09	Verification Status:	<<calculated Field: if Value=pass in all rows in Range D01-D07, Then display result=System Complies; elseif D08 has an explanation, then display result=System Complies; else display result=System Does Not Comply: one or more applicable remedial actions were not completed as required. Enter reason(s) for non-compliance in corrections notes field below>>
10	Correction Notes:	<<if Verification Status= Fail, then text entry in this Corrections Notes field is required as condition of registration of the failure; user input text>>
11	Optional Notes:	<<response is optional: user input text>>

E1. Forced Air System Airflow Rate Measurement

<<If A06 Condenser Speed Type != SingleSpeed Or A07_ResidentialCoolingZoneType != ZoneControlled this section is required; Else only display Section title and this note: *This section does not apply to this project* >>

Complete this section with Airflow Rate Measurement data for Newly Installed Non-Zoned Systems or Zoned Multi-Speed Compressor

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.

01	Actual System Airflow Rate Measurement (cfm)	<<user input numeric value: xxxx>>
02	Required Minimum System Airflow Rate (cfm/ton)	<<calculated field: If MCH-01 – ResidentialCoolingSystemType = Small Duct High Velocity AC or Small Duct High Velocity HP, then value = 250 Elseif, A04 = Alteration, then value = 300; Elseif parent is MCH-01a only: If (B10 ≠ NA and B10 ≠ D09), then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01a C08; Elseif parent is MCH-01d only: If (B08 ≠ NA and B08 ≠ D09) then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01d C08; Else value = 350>>
03	Required Minimum System Airflow Target (cfm)	<<calculated field: = A05 multiplied by value in E02>>
04	Compliance Statement:	<<If E01≥E03, result = true with display text "System airflow rate complies", else result = false If result is false Then display text "System does not comply with minimum airflow rate requirement">>

E2. Forced Air System Airflow Rate Measurement – All Zones Calling

If A06 CondenserSpeedType == SingleSpeed And A07_ResidentialCoolingZoneType== ZoneControlled this section is required Else only display Section title and this note: *This section does not apply to this project* >>

Complete this section with Airflow Rate Measurement data for Newly Installed Zoned Single-Speed Compressor Systems

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.

01	Actual System Airflow Rate Measurement (cfm)	<<user input numeric value: xxxx>>
02	Required All Zones Calling Minimum System Airflow Rate (cfm/ton)	<<calculated field: If MCH-01 – ResidentialCoolingSystemType = Small Duct High Velocity AC or Small Duct High Velocity HP, then value = 250 Elseif, A04 = Alteration, then value = 300; Elseif parent is MCH-01a only: If (B10 ≠ NA and B10 ≠ D09), then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01a C08; Elseif parent is MCH-01d only: If (B08 ≠ NA and B08 ≠ D09) then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01d C08; Else value = 350>>
03	Required All Zones Calling Minimum System Airflow Target (cfm)	<<calculated field: = A05 multiplied by value in E02>>
04	Compliance Statement:	<<If E01≥E03, result = true with display text "System airflow rate complies", else result = false If result is false Then display text "System does not comply with minimum airflow rate requirement">>

E3. Forced Air System Airflow Rate Measurement - Best Airflow Rate Attainable

If A04 = Alteration And A11 = RA3.3.3.1.5 this section is required; Else only display Section title and this note: *This section does not apply to this project >>*

Complete this section with Airflow Rate Measurement data for Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.

01	Actual System Airflow Rate Measurement (cfm)	<<user input numeric value: xxxx>>
02	Required Minimum System Airflow Rate (cfm/ton)	<<calculated field: If MCH-01 – ResidentialCoolingSystemType = Small Duct High Velocity AC or Small Duct High Velocity HP, then value = 250 Elseif, A04 = Alteration, then value = 300; Elseif parent is MCH-01a only: If (B10 ≠ NA and B10 ≠ D09), then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01a C08; Elseif parent is MCH-01d only: If (B08 ≠ NA and B08 ≠ D09) then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01d C08; Else value = 350>>
03	Required Minimum System Airflow Target (cfm)	<<calculated field: = A05 multiplied by value in E02>>
04	Compliance Statement:	<<If E01≥E03, result = true with display text "System airflow rate complies", else result = false If result is false Then display text " The measured Airflow Rate reported on this document is the best airflow rate attainable for compliance utilizing the procedures given in RA3.3.3.1.5. This system shall not be included in a sample group for ECC verification compliance >>
05	ECC Sample Group Eligibility	<<if E04 result = true with display text "System complies with minimum airflow rate requirement" then result = true display text "Eligible for ECC sample group"; else result = false display text "Not Eligible for ECC Sample Group for Airflow"

E4. Forced Air System Airflow Rate Measurement

If cooling system type on MCH-01 = No Cooling And A05 = N/A And A06 Condenser Speed Type != SingleSpeed Or

A07_ResidentialCoolingZoneType != ZoneControlled, this section is required; Else only display Section title and this note: *This section does not apply to this project >>*

Complete this section with Airflow Rate Measurement data for Newly Installed Heating Only Non-Zoned Systems or Zoned Multi-Speed Compressor Measurement Only - No Minimum Target Requirement

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.

01	Actual System Airflow Rate Measurement (cfm)	<<user input numeric value: xxxx>>
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F. Forced Air System Airflow Rate Measurement – All Other Zonal Control Modes

<< If A06 CondenserSpeedType == SingleSpeed And A07_ResidentialCoolingZoneType == ZoneControlled this section is required; Else only display Section title and this note: *This section does not apply to this project* >>

The procedures for System Airflow Rate Verification are specified in Reference Residential Appendix RA3.3.

For compliance with verification in all zonal control modes, it is sufficient to verify airflow rate for operation of each individual zone when the individual zone is the sole zone calling for conditioning. It is not necessary to verify airflow rate for combinations of 2 or more zones that are less than all zones calling (e.g., 2 out of three zones calling).

01	Number of Independently Controlled Zones (i.e., number of thermostats or temperature sensors that independently control one or more dampers.)	<<user input, integer>> (For each zone create a row in Table below: fields 04 through 07)		
02	Required Minimum Cooling System Airflow Rate (cfm/ton)	<<calculated field: If MCH-01 – ResidentialCoolingSystemType = Small Duct High Velocity AC or Small Duct High Velocity HP, then value = 250 Elseif, A04 = Alteration, then value = 300; Elseif parent is MCH-01a only: If (B10 ≠ NA and B10 ≠ D09), then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01a C08; Elseif parent is MCH-01d only: If (B08 ≠ NA and B08 ≠ D09) then value = 350; Elseif C08 ≠ NA, then reference value from MCH-01d C08; Else value = 350>>		
03	Required Minimum Airflow in all Zonal Control Modes (cfm)	<<calculated field: = A05 multiplied by value in F02>>		
	04	05	06	07
	Zone Name	Zone Description	Measured Airflow with All Other Zones Off (CFM)	Zone Compliance Status
	<<user entry, text field 20 characters>>	<<user entry, text field 50 characters>>	<<user input numeric value: xxxx>>	<<if F06 ≥ F03, then print “Pass”, else print “Fail”>>
08	Compliance Statement:		<<If all rows = “pass” in column F07, then display text "system airflow rate complies", else display text "system does not comply with minimum airflow rate requirement">>	

G. Central Fan Ventilation Cooling System Airflow Rate Measurement

<<If A12_CFVCS_Type == VariableCFVCS Or FixedCFVCS this section is required ; Else show Section Title and text: “This section does not apply to this project”>>

The procedures for central fan ventilation cooling system airflow rate verification are specified in Reference Residential Appendix RA3.3.4

01	Required Ventilation System Airflow Rate (cfm)	<<calculated field: If parent document = CF2R-MCH-01b, then value = 300; Else reference MCH-01 C11, Central Fan Ventilation Cooling Airflow>>
02	Actual System Ventilation Airflow Rate Measurement (cfm)	<<user input numeric value: xxxx>>
03	Compliance Statement:	<<If A12 = ‘Fixed CFVCS’, then if G02≥G01, result = "System ventilation airflow rate complies" Elseif A12 = ‘Variable CFVCS’, then if G02≤G01, result = "System ventilation airflow rate complies" Else display text "System does not comply with ventilation airflow rate requirement" >>

H1. Additional Requirements for All Systems

01	Air filters that meet the applicable requirements of Standards Section 150.0(m)12 or 150.0(m)13 were properly installed in the system during system airflow rate measurement identified on this Certificate of Installation.	
02	The airflow rate measurement apparatus used to perform the airflow rate measurement identified on this Certificate of Installation was calibrated in accordance with the apparatus manufacturer's specifications and conforms to the instrumentation specifications given in RA3.3.1.	
03	All registers were fully open during the diagnostic test.	
04	System fan was set at maximum speed during the diagnostic test.	
05	If fresh air duct is part of the HVAC system it was not closed during the diagnostic test.	
06	Airflow rate and fan watt draw shall be simultaneous measurements when used to calculate the Fan Efficacy tested value.	
07	Verification Status:	<<user pick from list: *** Pass - all applicable requirements are met; or *** Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or *** All n/a - This entire table is not applicable>>
08	Correction Notes:	<<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

H2. Additional Requirements for Systems with Heating And Cooling

01	A visual inspection shall confirm that bypass ducts that deliver conditioned supply air directly to the space conditioning system return duct airflow are not used on newly constructed zonally controlled systems unless the Performance Certificate of Compliance indicates an allowance for use of a bypass duct. When a bypass duct is accounted for on the Performance Certificate of Compliance, the airflow rate shall conform to the specifications listed on the Certificate of Compliance.	
02	Multi-speed compressor space cooling systems or variable speed compressor systems with controls that vary fan speed subject to the number of zones, as certified by the installer may verify airflow (cfm/ton) and fan efficacy (Watt/cfm) with system operating at maximum compressor capacity and system fan speed with all zones calling for conditioning,	
03	Verification Status:	<<user pick from list: *** Pass - all applicable requirements are met; or *** Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or *** All n/a - This entire table is not applicable>>
04	Correction Notes:	<<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

CERTIFICATE OF VERIFICATION – DATA FIELD DEFINITIONS AND CALCULATIONS	CF3R-MCH-23-H
Space Conditioning System Airflow Rate	(Page 8 of 8)

H3. Additional Requirements for Altered Systems Using Alternative Compliance		
01	For altered systems that do not comply with the minimum 300 cfm per ton airflow rate requirement but opt to comply using the remedial actions on this MCH-23 compliance document according to Section RA3.3.3.1.5, the system's thermostat shall conform to the specifications in Reference Joint Appendix JA5 and shall be capable of receiving and responding to Demand Response Signals prior to final approval of the building permit by the enforcing agency (Section 150.2(b)1Fiia).	
02	Verification Status:	<<user pick from list: *** Pass - all applicable requirements are met; or *** Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or *** All n/a - This entire table is not applicable>>
03	Correction Notes:	<<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.		

I. Determination of ECC Verification Compliance	
All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.	
01	<< If B01_HSPP_PSPP_ComplianceMethod != HSPP_PSPP_NotInstalled; and C04_IsAirflowMeasurementDeviceCertified == true; and (Ems04_DoesForcedAirSystemAirflowRateComply Or Essz04_DoesForcedAirSystemAirflowRateComply Or Ealt04_DoesForcedAirSystemAirflowRateComply == true) And If present D09_DoesForcedAirSystemPassAlternativeComplianceRequirements == true And If present F08_DoAllZonesComplyWithRequiredAirflowRate == true; And If present G03_DoesCentralFanVentilationCoolingSystemAirflowComply == true and H07_DoesSystemComplyWithRequirements == Pass or N/A And If present Hhc03_DoesSystemComplyWithRequirements == Pass or N/A And If present Halt02_DoesSystemComplyWithRequirements == Pass or N/A then result = true display term Complies: All specified verification protocol requirements on this document are met; Else result = false display term Does not comply: One or more specified verification protocol requirements on this document are not met.>>